Phase 2: Probability and Probability Distributions

Dietary supplement use and sleep in NHANES, August 2021-August 2023

Common tasks 2.1-2.9 and NHANES Path A requirements A1-A4

Analysis, tables, figures, and reproducibility details are provided in the following sections. The principal empirical probability sample contains 4,741 adults. Numeric sleep summaries and weighted comparisons use the documented outcome-specific subsets.

Phase 2.1. Events and Random Variable

Purpose

This section translates the project question into probability notation. The target population remains adults aged 20 years or older represented by NHANES August 2021–August 2023. Each observation is one participant, identified by SEQN.

Probability analysis set

For the initial empirical probability calculations, include participants who:

• have linked records in DEMO_L, SLQ_L, and DSQTOT_L;

• are aged 20 years or older (RIDAGEYR >= 20);

• have a valid yes/no response for any supplement use (DSD010 is 1 or 2); and

• have a non-missing weekday/workday sleep response (SLD012).

Using the downloaded files, this initial complete-observation set contains 4,741 adults. Each participant appears once. The unweighted empirical probabilities will use this set as their denominator. A separate weighted version will be computed using positive WTDRD1 values and reported as a population-oriented estimate; it will not be mixed with the unweighted sample proportions.

Events

For the unweighted empirical probability, treat each participant in the observed probability analysis set as equally likely to be selected. This describes the analyzed sample. The separate WTDRD1-weighted estimate is the population-oriented version and accounts for the NHANES selection weights.

• Event A — Any supplement use: the participant reported taking at least one dietary supplement in the past 30 days (DSD010 = 1). Its complement, A■, is a valid report of no supplement use (DSD010 = 2). Refused, “don’t know,” and missing responses are outside this analysis set rather than being coded as “no.”

• Event B — Weekday short sleep: the participant’s usual weekday/workday sleep duration is less than 7 hours. In SLD012, code 2 means “less than 3 hours,” so it belongs to B; exact reported values from 3 through 6.5 hours also belong to B. Values of 7 hours or more, including the grouped code 14 (“14 hours or more”), are not members of B (they belong to B■). Missing sleep responses are excluded from the analysis set.

The 7-hour cutoff is a pre-specified project definition for a binary probability event, informed by CDC guidance that adults generally need at least 7 hours of sleep. It is not a clinical diagnosis and does not imply that all adults have the same sleep needs. NHANES records the low and high sleep endpoints as grouped categories, so code 2 is classified as below 7 hours and code 14 as not below 7 hours; neither is treated as an exact number of hours.

Random variable

Let X be the participant’s usual weekday/workday sleep duration, measured in hours and recorded in SLD012. X is a quantitative random variable with half-hour increments for exact values. Because the released codes 2 and 14 represent ranges, analyses of X’s numeric expectation and variance must state how those grouped endpoints are handled. For the binary event B, their categories are sufficient to determine whether sleep is below 7 hours.

For subgroup comparisons, the event B will be examined across the pre-specified age bands (20–39, 40–59, and 60+) and NHANES sex categories (RIAGENDR 1 and 2). These comparisons are descriptive probability calculations; Phase 2 does not perform formal hypothesis tests.

Link to the project question

The central probability comparisons will include P(B | A) and P(B | A■): the empirical probability of weekday short sleep among supplement users and among non-users. Other quantities in the next section can include the marginal probabilities P(A) and P(B), their joint probability P(A ∩ B), and conditional probabilities within the pre-specified age or sex groups. Differences will be interpreted descriptively and will not be described as causal effects.

Data documentation

- NHANES sleep questionnaire documentation (SLQ_L)
- NHANES total supplement-use documentation (DSQTOT_L)
- NHANES demographics documentation (DEMO_L)
- CDC adult sleep guidance

02-Marginal and Joint Empirical Probabilities

Analysis set

This calculation uses the 4,741 adults defined in Task 1: linked NHANES records, age 20 years or older, a valid yes/no response for DSD010, and a non-missing response for weekday sleep (SLD012). These are unweighted empirical sample probabilities. The sampling-weighted comparison is reserved for the NHANES-specific Path A requirement.

The two events are:

- A: any dietary supplement use in the past 30 days (DSD010 = 1); A■: no use (DSD010 = 2).
- B: usual weekday/workday sleep under 7 hours (SLD012 < 7, including the grouped <3 code).

Marginal probabilities

For an event E, the empirical probability is its observed frequency divided by the number of eligible observations, n = 4,741.

Event	Count	Empirical probability	Percent
A: any supplement use	3,070	3,070 / 4,741 = 0.6475	64.75%
A■: no supplement use	1,671	1,671 / 4,741 = 0.3525	35.25%
B: weekday sleep under 7 hours	965	965 / 4,741 = 0.2035	20.35%
B■: weekday sleep 7 hours or more	3,776	3,776 / 4,741 = 0.7965	79.65%

Joint probabilities

	B: under 7 hours	B■: 7 hours or more	Row total
A: any supplement use	546 (11.52%)	2,524 (53.24%)	3,070 (64.75%)
A■: no supplement use	419 (8.84%)	1,252 (26.41%)	1,671 (35.25%)

	B: under 7 hours	B■: 7 hours or more	Row total
Column total	965 (20.35%)	3,776 (79.65%)	4,741 (100.00%)

For example, `P(A ∩ B) = 546 / 4,741 = 0.1152`. Each joint-cell percentage uses the full analysis-set denominator. The four joint probabilities sum to 1 before rounding, and their row and column totals agree with the marginal probabilities above. Displayed percentages may differ from 100% by 0.01 percentage point because of rounding.

Interpretation

In this complete-observation adult sample, about 64.75% reported any supplement use, and about 20.35% were classified as usually sleeping under 7 hours on weekdays. About 11.52% of the sample met both event definitions. These are sample descriptions; they are not yet population estimates and do not show that supplement use changes sleep. NHANES weighting and conditional probabilities are handled in the following tasks.

Reproducibility

Run `python src/phase2_analysis.py` from the project root. The script reads the original `DEMO_L.xpt`, `SLQ_L.xpt`, and `DSQTOT_L.xpt` files, checks that `SEQN` is unique within each participant-level file, recreates the adult analysis set, and writes the marginal and joint tables to `data/processed/phase2/`. The original XPT files remain unchanged.

Data documentation

- NHANES sleep questionnaire documentation (`SLQ_L`)
- NHANES total supplement-use documentation (`DSQTOT_L`)

03-Conditional Probabilities

Question

Among adults in the probability analysis set, how common is weekday sleep under 7 hours among participants who reported any supplement use, and how common is it among participants who reported no supplement use?

Let:

- A = reported any supplement use in the past 30 days.
- A^c = reported no supplement use.
- B = usual weekday/workday sleep under 7 hours.

The calculation uses the same 4,741 adults defined in Task 1. These are unweighted empirical probabilities.

Calculations

The conditional probability formula is:

$P(B \mid A) = P(A \cap B) / P(A)$

For supplement users, 546 of the 3,070 participants with A also belonged to B:

$P(B \mid A) = 546 / 3{,}070 = 0.1779 = 17.79\%$

For participants reporting no supplement use, 419 of the 1,671 participants with A^c belonged to B:

$P(B \mid A^c) = 419 / 1{,}671 = 0.2507 = 25.07\%$

Supplement-use group	Participants in group	Under 7 hours	Conditional probability
Any supplement use (A)	3,070	546	17.79%
No supplement use (A^c)	1,671	419	25.07%

The observed difference is 17.79% − 25.07% = −7.29 percentage points (rounded from unrounded values). In this analyzed sample, the proportion classified as usually sleeping under 7 hours is descriptively lower among participants reporting supplement use.

Why the denominator changes

A conditional probability restricts attention to the group named after the vertical bar. For P(B | A), the denominator is the 3,070 supplement users, not all 4,741 adults. For $P(B \mid A^c)$, it is the 1,671 non-users. The numerators count the people in each restricted group who also meet B. This is why these conditional percentages answer a different question from the joint percentages in Task 2, which used all 4,741 adults as their denominator.

Interpretation and limits

These are descriptive probabilities for the complete-observation sample. The difference does not establish that supplement use affects sleep: the data are cross-sectional, supplement use combines many products, and groups may differ in age or other characteristics. No significance test or causal interpretation is made here. Sampling-weighted versions and subgroup conditional probabilities are addressed in the NHANES-specific Path A requirements.

Reproducibility

Run `python src/phase2_analysis.py` from the project root using the project environment. The script calculates these probabilities from the source XPT files and writes `conditional_probabilities.csv` under `data/processed/phase2/`. Raw XPT files are not modified.

Data documentation

- NHANES sleep questionnaire documentation (`SLQ_L`)
- NHANES total supplement-use documentation (`DSQTOT_L`)

04-Independence of Events

Question and analysis set

Are the events of reporting any dietary supplement use and usually sleeping under 7 hours on weekdays independent in the empirical distribution of the analyzed adults?

Use the same 4,741 adults and event definitions as in Task 1: A means any supplement use in the past 30 days (`DSD010 = 1`), and B means usual weekday/workday sleep under 7 hours (`SLD012 < 7`). The analysis includes adults age 20 or older with valid values for both variables. All probabilities below are unweighted sample proportions.

Criterion for independence

Two events are independent when knowing whether one occurred leaves the probability of the other unchanged. One equivalent criterion is P(A ∩ B) = P(A) × P(B). We compare the observed joint proportion with the product of the two observed marginal proportions. Exact equality is the independence condition for a probability distribution; a finite sample can depart from equality through sampling variation.

Empirical comparison

Quantity	Calculation	Empirical value
P(A): any supplement use	3,070 / 4,741	0.6475 (64.75%)
P(B): weekday sleep under 7 hours	965 / 4,741	0.2035 (20.35%)
Observed P(A ∩ B): both events	546 / 4,741	0.1152 (11.52%)
P(A) × P(B): joint probability under independence	(3,070 / 4,741) × (965 / 4,741)	0.1318 (13.18%)

The observed joint proportion is 1.66 percentage points lower than the product of the marginal proportions (`0.1152 − 0.1318 ≈ −0.0166`). The unrounded calculations are stored in `data/processed/phase2/independence_comparison.csv`.

The conditional probabilities in Task 3 give the same picture: P(B | A) = 546 / 3,070 = 17.79%, while P(B | A^c) = 419 / 1,671 = 25.07%. If A and B were independent in this empirical distribution, these conditional proportions would equal the overall P(B) of 20.35%.

Interpretation and limits

The empirical distribution of this analyzed sample does not satisfy the independence equality. Short weekday sleep is less common among respondents reporting any supplement use than among respondents reporting no supplement use. This is a descriptive sample association. Sampling variation, differences between the groups, and self-reported measurements all matter when interpreting it. A conclusion about independence in the wider population would require survey-design-aware inference, which is outside this descriptive task. The cross-sectional comparison does not show that supplements cause a change in sleep.

Reproducibility

Run `python src/phase2_analysis.py` from the project root with the project environment active. The script reads the original XPT files, recreates the complete-observation adult sample, and writes `data/processed/phase2/independence_comparison.csv`. The marginal and joint counts in Task 2 provide a separate check of the calculations.

Phase 2.5. Discrete Count and the Binomial Model

Count and sampling experiment

Let Y be the number of adults who usually sleep under 7 hours on weekdays among 10 distinct supplement users selected uniformly from the 3,070 supplement users in the Phase 2 analysis set. Each selected person contributes one of two outcomes: short weekday sleep (`SLD012 < 7`) or not. Thus Y is a discrete count with possible values 0, 1, …, 10. In the fixed analyzed sample, 546 of the 3,070 supplement users meet the short-sleep definition, giving an empirical success fraction of p = 546 / 3,070 = 0.17785.

This is a clearly defined *resampling illustration* using the analyzed sample. It is separate from the count of 546 short sleepers observed among all 3,070 supplement users in NHANES.

When a binomial model applies

A binomial model for Y needs a fixed number of trials, two outcomes per trial, independent trials, and the same success probability on every trial. If the 10 users were drawn with replacement from the fixed group, each draw would have probability p of selecting a short sleeper and the draws would be independent. Conditional on that fixed group, the count would follow `Binomial(10, 546/3070)` exactly.

Our defined experiment selects distinct people without replacement. Its draws are therefore dependent: after a short sleeper is selected, both the number of short sleepers and the total number of people left decrease. The exact distribution is hypergeometric, with 3,070 people, 546 short sleepers, and 10 draws. A binomial distribution with `n = 10` and `p = 546/3070` is a close approximation here because 10 / 3,070 is only 0.33% of the group.

For example, for exactly two short sleepers:

Model	Calculation	P(Y = 2)
Binomial approximation	$C(10, 2) \times p^2 \times (1 - p)$■	0.297118 (29.71%)
Exact sampling without replacement	`C(546, 2) × C(2524, 8) / C(3070, 10)`	0.297634 (29.76%)

The two probabilities differ by about 0.05 percentage point. The script calculates the complete probability distribution for all counts from 0 through 10; this example shows how closely the two models agree for one outcome. The binomial approximation is based on the observed empirical fraction p, not a known population probability.

Why the NHANES count is a different problem

The actual 546 short sleepers among the 3,070 analyzed supplement users should not be treated as the outcome of 3,070 independent, equal-probability Bernoulli trials. NHANES uses a stratified, multistage sample with unequal selection probabilities and clustering. Respondents also vary in characteristics related to sleep, and inclusion in this complete-observation subset depends on having valid supplement and sleep responses. Those features undermine the independent, common-p assumptions for a simple binomial model of the survey count. Applying a simple binomial standard error to that count would ignore the survey design.

This section is a probability-model illustration. It makes no significance test, population estimate, or causal claim about supplement use and sleep.

Reproducibility

Run `python src/phase2_analysis.py` from the project root using the project environment. The script rebuilds the analysis set from the original XPT files and writes `data/processed/phase2/binomial_illustration.csv`, with the binomial approximation and exact without-replacement probability for each count from 0 through 10. The original files remain unchanged.

Phase 2.6. Quantitative Sleep Duration and the Normal Model

Variable and analysis set

The underlying quantity is an adult's usual weekday/workday sleep duration in hours. NHANES derives `SLD012` from reported sleep and wake times and rounds it to the nearest half hour. The released variable is therefore discretized, even though duration is conceptually continuous. Code `2` means *less than 3 hours* and code `14` means *14 hours or more*; neither is an exact duration. For the numeric distribution and normal comparison, retain values from 3 through 13.5 hours.

The Phase 2 analysis set contains 4,741 adults with valid supplement-use and weekday-sleep responses. Of these, 4,702 have exact-hour codes and 39 have one of the grouped endpoint codes. The histogram and normal Q-Q comparison below use the 4,702 exact-hour records, unweighted.

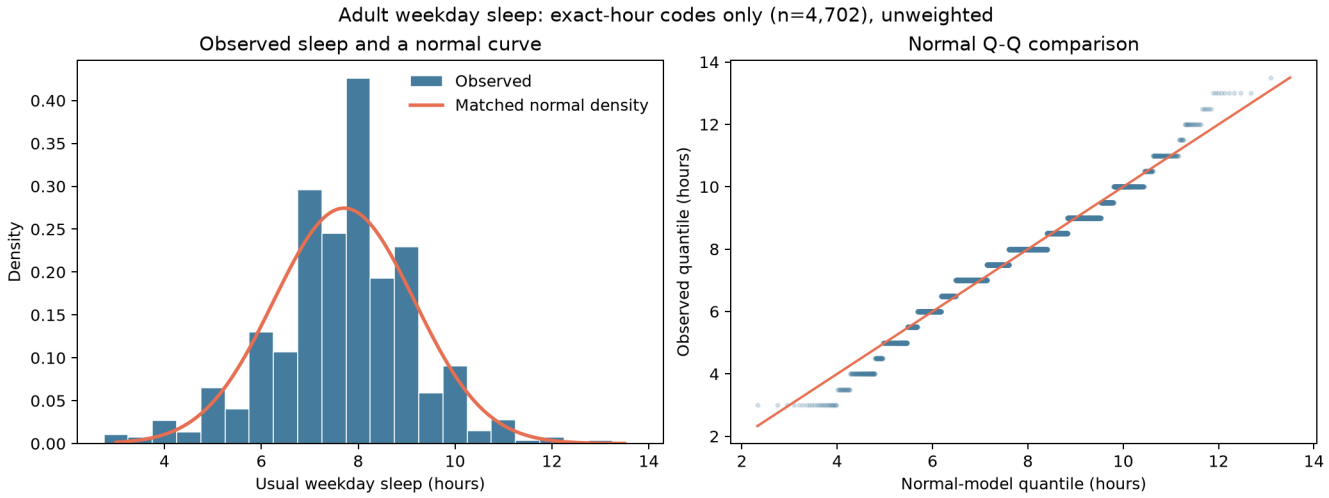


Figure 2. Histogram with a normal curve matched to the empirical mean and standard deviation, and a normal Q-Q comparison.

Shape and normal approximation

The observed mean is 7.71 hours, the median is 8.0 hours, and the standard deviation is 1.45 hours. The sample skewness is -0.18, indicating mild left skew by this measure. The histogram has a strong concentration at round and half-hour values, especially around 8 hours. The Q-Q plot shows steps caused by rounding and departures from the reference line in the tails. About 74.84% of exact-hour values fall within one empirical standard deviation of the mean, compared with about 68.27% for an exact normal distribution.

A fitted normal curve gives a rough description of the center but is not a faithful model of the observed duration distribution. It smooths over the half-hour heaping, does not reflect the grouped endpoint categories, and misses features of the tails. No normality test is used as a substitute for checking the plotted shape.

Reproducibility and source

Run `python src/phase2_extended_analysis.py` from the project root. The script writes `data/processed/phase2/weekday_sleep_distribution_summary.csv` and the figure above. See the NHANES sleep questionnaire documentation for the derivation, rounding, and grouped endpoint codes.

Phase 2.7. Expectation and Variance of Weekday Sleep Duration

Empirical distribution

Let X be usual weekday/workday sleep duration for a uniformly selected adult from the 4,702 exact-hour records described in Task 2.6. Each observed person receives probability `1/4702` in this unweighted empirical distribution. Values coded as *less than 3 hours* or *14 hours or more* are excluded because their exact hours are unknown.

Quantity	Definition for the empirical distribution	Result
E[X], empirical mean	Sum of the 4,702 exact durations / 4,702	7.7084 hours
Var(X), empirical variance	Mean of $(\texttt{duration} - 7.7084)^2$ over the 4,702 records	2.1130 hours²
SD(X)	Square root of empirical variance	1.4536 hours
Median and interquartile range	Median; Q3 - Q1	8.0 hours; 1.5 hours

The expectation of 7.71 hours is the average weekday duration in this empirical distribution. The variance measures average squared distance from that mean; its squared-hour units make it less directly readable than the standard deviation of 1.45 hours. The median and IQR supplement the mean and variance because the recorded durations are heaped at half hours and the distribution is not exactly normal.

These calculations use denominator n, rather than `n - 1`, because they describe the observed empirical distribution. They are not an estimate of the design-based population variance or the sampling variance of a survey estimate. The mean and variance are sensitive to how grouped endpoints are handled, so the exclusion rule must accompany the result.

Reproducibility

Run `python src/phase2_extended_analysis.py` from the project root. The calculations and sample counts are written to `data/processed/phase2/weekday_sleep_distribution_summary.csv`.

Phase 2.8. Two Sleep Variables and Covariance

Paired variables

For the same adult, let X be usual weekday/workday sleep hours (`SLD012`) and Z be usual weekend/non-workday sleep hours (`SLD013`). Both measures must be exact-hour codes from 3 through 13.5 hours. The paired analysis has 4,669 adults, each contributing one pair. This is a separate complete-pair subset of the Phase 2 analysis set, and all calculations are unweighted.

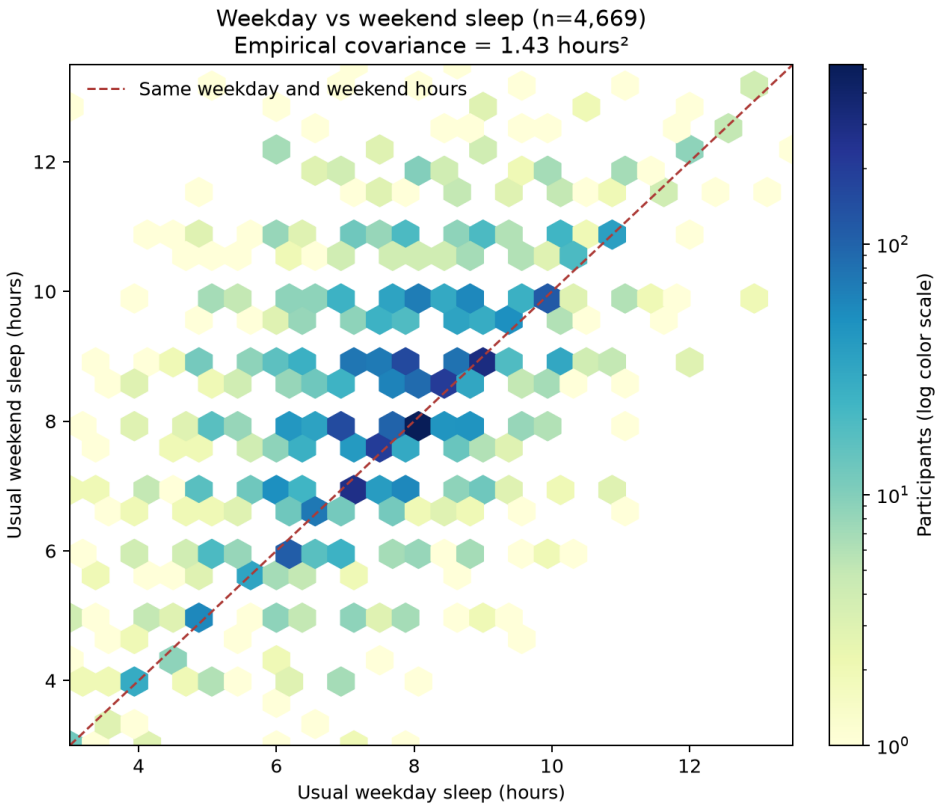


Figure 3. Hexbin scatter display of paired weekday and weekend sleep hours; darker cells contain more adults.

Empirical covariance and interpretation

The paired-sample means are 7.7081 hours on weekdays and 8.2269 hours on weekends. Using the empirical-distribution denominator $n = 4,669$, the covariance is

$$\mathrm{Cov}(X, Z) = \mathrm{mean}[(X - \mathrm{mean}(X)) \times (Z - \mathrm{mean}(Z))] = +1.4279 \text{ hours}^2.$$

The positive sign means that adults with above-average weekday sleep tend, in this sample, also to have above-average weekend sleep. The plot shows the upward association and the concentration at rounded half-hour values. Many observations lie above the equal-hours line, consistent with a higher average weekend duration in this paired subset. Covariance depends on the units and spread of both variables, so its numerical size is not a standardized strength measure. Formal correlation and population inference are reserved for a later phase.

The two durations are paired observations from the same person. Their covariance describes co-variation; it does not show that one duration causes the other or that supplement use explains the relationship.

Reproducibility

Run `python src/phase2_extended_analysis.py` from the project root. The paired count, means, and covariance are written to `data/processed/phase2/weekday_weekend_covariance.csv`.

Phase 2.9. Bayes' Rule for Supplement Use and Short Sleep

Updating the direction of the question

The previous conditional calculation asked for the probability of short weekday sleep given supplement use, `P(B | A)`. Here the question is reversed: among adults who usually sleep under 7 hours on weekdays, what fraction reported any dietary supplement use in the past 30 days? With `A` = supplement use and `B` = short weekday sleep, this is `P(A | B)`.

In the same 4,741-adult unweighted analysis set, `P(A) = 3,070/4,741 = 0.64754`, `P(B | A) = 546/3,070 = 0.17785`, and `P(B) = 965/4,741 = 0.20354`. Bayes' rule gives:

`P(A | B) = P(B | A) × P(A) / P(B)`

`= (546/3070) × (3070/4741) / (965/4741) = 546/965 = 0.5658`.

Thus 56.58% of the 965 short-sleep adults reported supplement use. For comparison, `P(A | B■) = 2,524/3,776 = 66.84%`. The overall supplement-use fraction before conditioning on sleep is 64.75%. The changed denominator explains why `P(A | B)` differs from `P(B | A) = 17.79%`.

Bayes' rule reorganizes the same observed joint and marginal frequencies; it does not add new evidence or make a causal claim. These are empirical sample probabilities, and the survey-weighted comparison is handled in Path A4.

Reproducibility

Run `python src/phase2_extended_analysis.py` from the project root. The calculation is written to `data/processed/phase2/bayes_reverse_conditional.csv`; the underlying counts can also be checked in Task 2.2.

Path A1. Conditional Short-Sleep Probabilities in Subgroups

Definition

The binary outcome is usual weekday/workday sleep under 7 hours (`B`, derived from `SLD012`). Within each pre-specified age band and NHANES sex category, compare `P(B | A, subgroup)` with `P(B | A■, subgroup)`, where `A` means reported any dietary supplement use in the past 30 days. The analysis uses the 4,741-adult Phase 2 sample, with one person per row. These are unweighted, descriptive sample proportions.

Age-band comparisons

Age band	Supplement use	Adults	Short-sleep adults	Conditional probability
20-39	Yes	522	92	17.62%
20-39	No	588	135	22.96%
40-59	Yes	807	165	20.45%
40-59	No	512	142	27.73%
60+	Yes	1,741	289	16.60%

Age band	Supplement use	Adults	Short-sleep adults	Conditional probability
60+	No	571	142	24.87%

The sample short-sleep fraction is lower among supplement users in each age band. For example, among adults aged 40-59, it is 20.45% for users and 27.73% for non-users, a descriptive difference of -7.29 percentage points. These within-band percentages address a more focused question than the overall comparison, because the age composition of the two supplement groups differs.

NHANES sex-category comparisons

`RIAGENDR` category	Supplement use	Adults	Short-sleep adults	Conditional probability
Male (code 1)	Yes	1,203	232	19.29%
Male (code 1)	No	881	237	26.90%
Female (code 2)	Yes	1,867	314	16.82%
Female (code 2)	No	790	182	23.04%

The same descriptive direction appears in both released sex categories. These comparisons are not tests of a supplement effect: the data are observational, supplement use is broad, other characteristics may differ, and survey weights are not used in this table. No hypothesis test is performed in Phase 2.

Reproducibility

Run `python src/phase2_extended_analysis.py` from the project root. The full subgroup counts and proportions are saved in `data/processed/phase2/path_a1_subgroup_conditional_probabilities.csv`. Age bands were defined in Task 2.1; `RIAGENDR` codes come from the NHANES demographics documentation.

Path A2. Binomial Counts in Equal-Size Subsamples

Construction

Use the 3,070 supplement users from the Phase 2 analysis set. Of them, 546 report usual weekday sleep under 7 hours, so the empirical short-sleep fraction is `p = 546/3070 = 0.17785`. With a fixed random seed (`20261113`), randomly permute the 3,070 users and partition them into 307 non-overlapping groups of 10. No user is left out. For each group, count the members with short weekday sleep. Compare the observed distribution of these 307 counts with a `Binomial(10, p)` model.

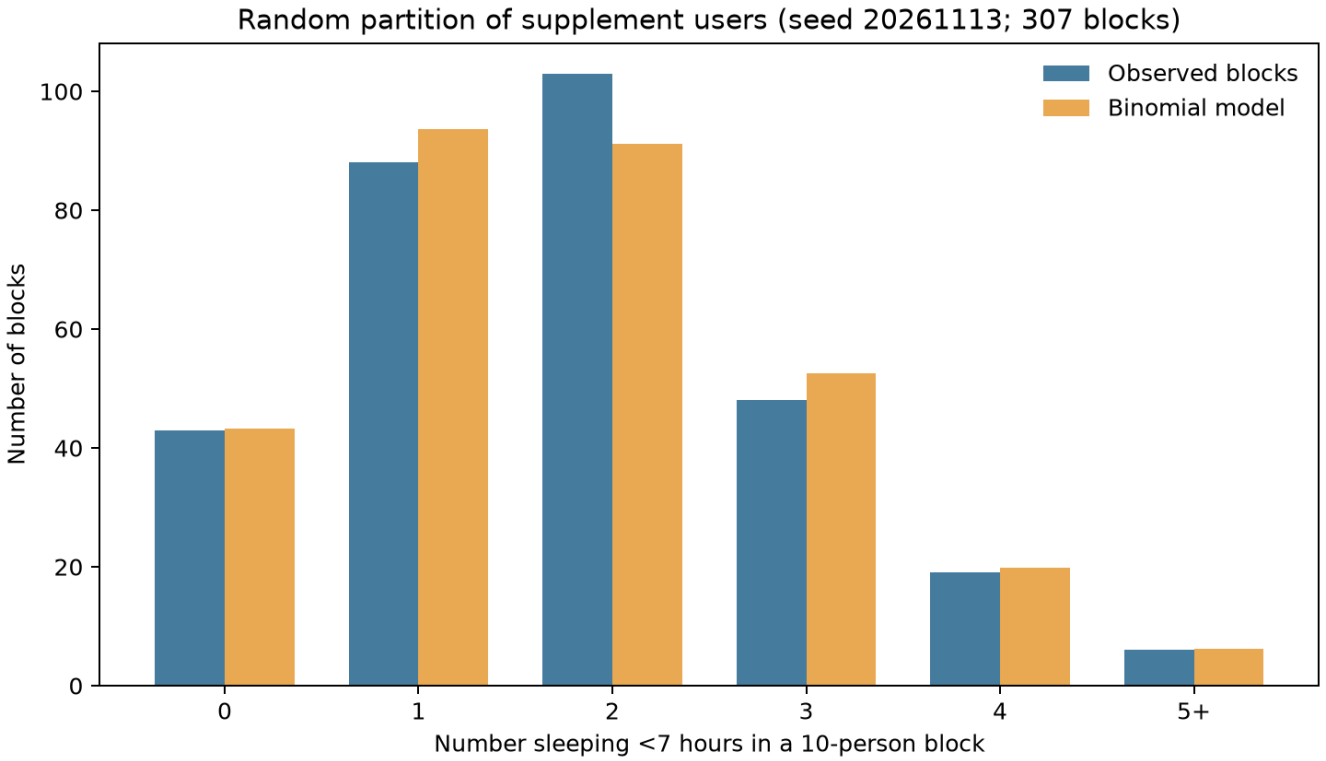


Figure 1. Observed counts in equal-size random groups compared with the binomial model; the sparse right tail is combined as 5 or more.

Short sleepers in a group of 10	Observed groups	Expected groups under `Binomial(10, 0.17785)`
0	43	43.3
1	88	93.7
2	103	91.2
3	48	52.6
4	19	19.9
5 or more	6	6.2
Total	307	307.0

The observed block-count variance is 1.430, compared with the binomial value `10p(1-p) = 1.462`. The observed mean and binomial mean are both 1.779 because every supplement user appears exactly once and p was calculated from those same users; that equality is built into this construction, not independent evidence of model fit. The observed and model frequencies are fairly close in this one random partition, while individual bins still differ.

Model limits

The groups were created after data collection by random partition; they are not NHANES sampling clusters. The counts in non-overlapping groups are constrained by the fixed total of 546 short sleepers, so the groups are not independent binomial samples. Within each group, selection is without replacement; Task 2.5 shows why binomial probabilities are still numerically close for ten draws from 3,070 users. Another random seed would give a different observed table. This descriptive comparison does not validate the binomial assumptions for population inference, and no goodness-of-fit test is claimed.

Reproducibility

Run `python src/phase2_extended_analysis.py` from the project root. The full observed and expected frequencies for counts 0 through 10 are saved to `data/processed/phase2/path_a2_equal_size_blocks.csv`.

Path A3. Shape of a Health-Related Measurement

Selected measure

The health-related measurement is usual weekday/workday sleep duration, derived from sleep and wake times in `SLD012`. The released values are rounded to half hours, and the two endpoint codes are grouped categories. For numerical shape analysis, use the 4,702 exact-hour values from 3 to 13.5 hours within the Phase 2 adult sample. The remaining 39 endpoint-coded records are kept for the binary short-sleep event but do not enter the numerical duration summary.

The histogram and normal Q-Q display are shown in Task 2.6. The mean is 7.71 hours, median 8.0 hours, and sample skewness -0.18. This indicates mild negative skew by a numeric measure, although the most striking shape feature is concentration at round and half-hour values. The distribution is unimodal around 7-8 hours and does not follow a smooth normal curve closely.

Unusual values and normal approximation

The first and third quartiles are 7.0 and 8.5 hours, so the IQR is 1.5 hours. A descriptive Tukey rule places fences at 4.75 and 10.75 hours. There are 137 exact-hour observations below the lower fence and 106 above the upper fence, for 243 flagged observations in total. The observed exact-hour range is 3.0 to 13.5 hours. These flags identify unusual values under this rule; they are not automatically errors and are not deleted. The 39 grouped endpoint codes require separate handling because their precise durations are undisclosed.

The normal curve misses the heaping and parts of both tails, as the Q-Q plot shows. It is only a rough approximation for the center and should not be assumed for the individual-level duration distribution. These are unweighted sample descriptions, not population estimates or clinical diagnoses.

Reproducibility and source

Run `python src/phase2_extended_analysis.py` from the project root. Results are in `data/processed/phase2/weekday_sleep_distribution_summary.csv`. The NHANES sleep documentation explains the derived values, half-hour rounding, and endpoint categories.

Path A4. Raw and WTDRD1-Weighted Empirical Probability

Weight and common comparison set

The main exposure, 30-day dietary supplement use (`DSD010`), comes from `DSQTOT_L`. Its NHANES analytic notes specify `WTDRD1`, the Day 1 dietary weight, for analysis involving the 30-day supplement data. This weight accounts for unequal selection probabilities and additional nonresponse and dietary-interview allocation. The linked demographics file also supplies the masked design variables `SDMVSTRA` and `SDMVPSU`.

For a fair raw-versus-weighted comparison, use the same 4,737 adults with valid supplement and weekday-sleep responses and a positive, finite `WTDRD1`. Four of the 4,741 adults in the earlier unweighted probability set are excluded here because their weight is not positive and finite. The outcome is `B`: usual weekday/workday sleep under 7 hours; endpoint code 2 (*less than 3 hours*) correctly counts as B.

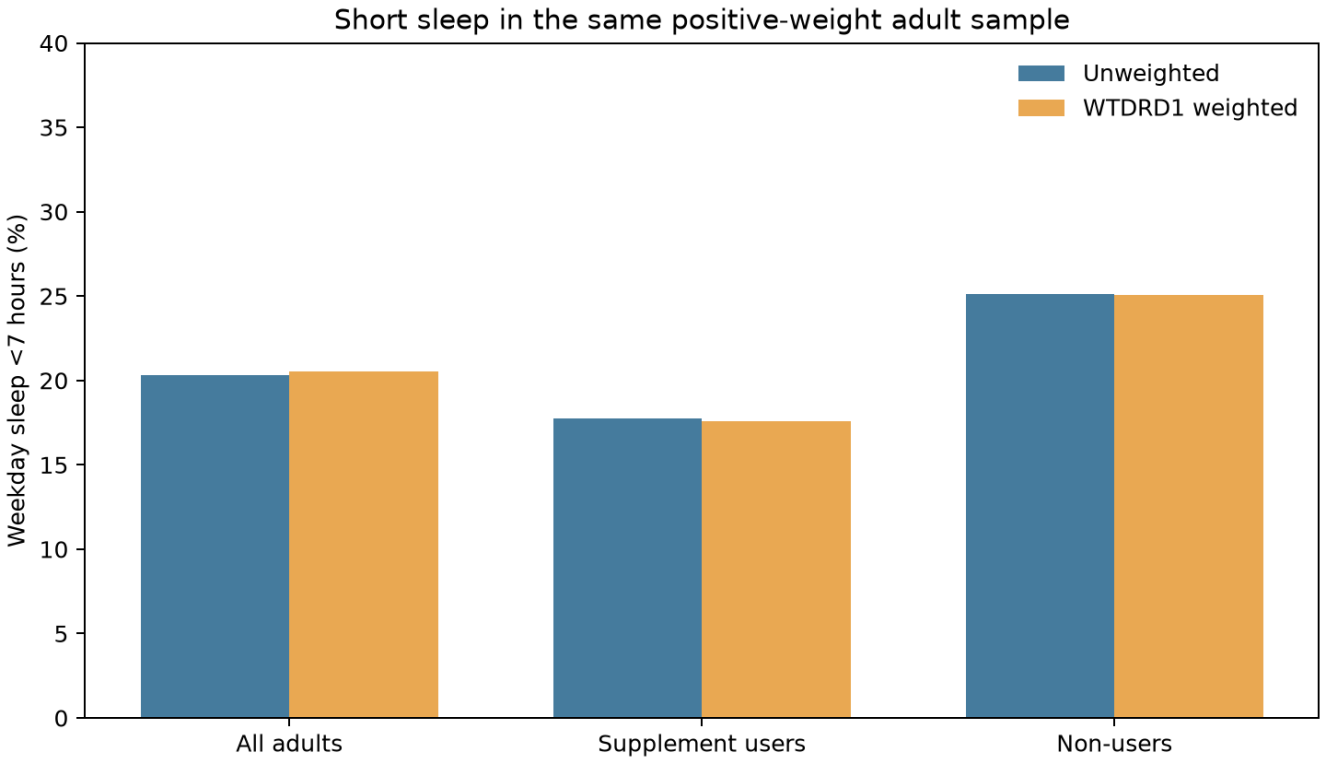


Figure 4. Unweighted and WTDRD1-weighted short-sleep proportions in the same positive-weight adult sample.

Group	Adults	Short-sleep adults	Unweighted P(B)	`WTDRD1`-weighted P(B)
All adults	4,737	963	20.33%	20.54%
Supplement users	3,068	544	17.73%	17.57%
Non-users	1,669	419	25.10%	25.04%

For each row, the raw proportion is `number with B / number in group`. The weighted proportion is `sum(WTDRD1 × 1[B]) / sum(WTDRD1)` over the same group. For all adults, weighting changes the point estimate by +0.21 percentage point (using the displayed rounding). The within-group changes are also small in this particular complete-case sample; their size must be read from the data rather than assumed.

An unweighted sample frequency gives every retained respondent equal influence. NHANES participants represent different numbers of people after selection and response adjustments, so a weighted ratio has a different population-oriented interpretation. A weight-only point estimate does not supply a design-based standard error or confidence interval: those also require appropriate use of `SDMVSTRA` and `SDMVPSU`. Complete-case selection and self-reported sleep may still affect interpretation. No significance test is performed in Phase 2.

Reproducibility

Run `python src/phase2_extended_analysis.py` from the project root. The shared-sample counts and both probability versions are saved to `data/processed/phase2/path_a4_weighted_probability.csv`. Phase 1's weight documentation records the rationale for choosing `WTDRD1`.